Roll No. | | | | | | | |

MID SEMESTER EXAMINATION MARCH - 2025

Name of the Program: **B.Tech., ME** Semester: **4th**

Name of the Course: **Automotive Materials and Manufacturing Techniques**

Course Code: **TME412**

Time: 90 Minutes Maximum Marks: 50

Note:-

(i) Answer **all the questions** by choosing **anyone of the sub questions**.

(ii) Each question carries 10 marks

Q1	(10marks)	**CO1**
(a)	Define ferrous and non-ferrous materials and provide two examples of each used in automotive applications.	
	OR	
(b)	List the key differences between high-strength steel (HSS) and advanced high-strength steel (AHSS) in terms of microstructure and mechanical properties.	
Q2	(10 marks)	**CO2**
(a)	Explain how the mechanical properties of high-strength steel (HSS) differ from those of conventional steels and why it is preferred in vehicle structures.	
	OR	
(b)	Explain why lightweight materials such as magnesium and carbon fiber-reinforced polymers (CFRP) are gaining popularity in automobile manufacturing.	
Q3	(10 marks)	**CO3**
(a)	A car manufacturer wants to replace some metal components with polymer-based materials. What factors should be considered while selecting polymers for automotive applications?	
	OR	
(b)	Describe how the forging process enhances the mechanical properties of automotive components compared to casting. Provide one example of a forged component.	
Q4	(10 marks)	**CO4**
(a)	Compare the advantages and limitations of composite materials over traditional metallic materials in automotive applications.	
	OR	
(b)	A manufacturer is considering casting and machining for producing engine blocks. What are the critical process selection factors that should be considered?	
Q5	(10 marks)	**CO5**
(a)	If you were an automobile design engineer, how would you optimize the use of aluminum alloys in an automotive chassis to reduce weight without compromising safety?	
	OR	
(b)	Evaluate the potential impact of additive manufacturing (3D printing) on automotive material selection and component manufacturing.	